

# LUMEN Quest - 2025 - Student Case Study

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## Telecom Inventory Management System

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# Telecom Inventory Management System

## Use Case Description

Design and develop a Telecom Inventory Management System (TIMS) that enables seamless management of telecom products and services inventory with features to manage users, products, suppliers, stock levels, and notifications.

## Technical Requirements

### 1. Frontend:

- Build using modern frameworks like React, Angular, or Vue.js.
- Responsive design to ensure usability on desktops, tablets, and mobile devices.

### 2. Backend:

- Implement RESTful APIs using Node.js, Django, Flask, or similar frameworks.
- Secure endpoints with authentication and authorization (e.g., JWT or OAuth2).

### 3. Database:

- Use a relational database (e.g., MySQL, PostgreSQL) with normalized tables:
  - Users (Roles, credentials).
  - Products (Details, stock levels, reorder points).
  - Suppliers (Contact information, order history).
  - Transactions (Stock in/out, timestamps).

### 4. Alerts:

- Set up notification triggers for stock levels and order delays using CRON jobs or event-driven notifications.

### 5. Data Handling:

- Input validation for fields like product codes, stock counts, and supplier information.
- Ensure secure handling of sensitive user and supplier data.

## Technologies

These are general guidelines and students are encouraged to explore beyond the technologies listed below.

- **Programming Languages**
  - Python
  - JavaScript/TypeScript
  - Java, C++, or C#
- **Frontend Development**
  - Frameworks like React, Angular, or Vue.js.
- **Backend Development**
  - Node.js, Django, or Flask for API
  - Spring Boot & Java
  - Ruby on Rails
  - AI/Machine Learning - Scikit-learn or TensorFlow, Random forest, or XGB/GBM or regression algos.
- **Databases**
  - MySQL, PostgreSQL, or MongoDB for data storage.

## Acceptance Criteria

### Minimum Viable Product (MVP) Features

#### 1. User Authentication and Role-Based Access

- Users can log in with a secure authentication mechanism.
- Roles (Admin, Manager, Staff) determine access rights:
  - Admin can manage users, products, suppliers, and stock.
  - Manager can view/edit products, suppliers, and manage stock transactions.
  - Staff can view product stock and perform stock transactions.

## 2. Product Management

- Users can add, edit, or delete product details with mandatory fields (e.g., name, category, stock level).
- Product categories are maintained and searchable.
- Stock in/out transactions can be recorded with manual input.
- Reorder points can be set for each product.

## 3. Supplier Management

- Admin can add, edit, or delete supplier details, including contact information.
- Order history and current order status can be recorded for each supplier.

## 4. Alerts and Notifications

- Notifications are triggered for low stock levels.
- Alerts are sent for pending or overdue orders.

## 5. Search and Filtering

- Users can search for products by name, category, or stock level.
- Filters allow narrowing results by stock status (e.g., low stock, out of stock).

# Nice-to-Have Features

## 1. Demand Forecasting

- The system uses historical stock data to predict future demand.
- Forecasting accuracy is tested with at least 80% reliability based on sample data.

## 2. Data Import/Export

- Users can import product data in CSV or Excel format to update inventory.
- Inventory data can be exported in CSV or Excel for reporting.

## 3. Reporting Dashboard

- A visual dashboard displays key metrics:
  - Current stock levels.
  - Low stock alerts.
  - Supplier order status summary.

**Recommended Data Set**

**Refer the attached Inventory\_DataSet.xlsx in email.**

| Evaluation Criteria            | Weightage % |
|--------------------------------|-------------|
| Functionality and Features     | 30%         |
| Technical Implementation       | 20%         |
| User Interface and Experience  | 20%         |
| Innovation and Creativity      | 15%         |
| Presentation and Documentation | 10%         |
| Team Collaboration             | 5%          |
| Total                          | 100%        |